

TECHNICAL PORTFOLIO DOSSIER

STRUCTURAL ENGINEERING · BIM COORDINATION · GEOTECHNICAL
SOLUTIONS

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Executive Engineering Statement

This dossier provides the engineering rationale, structural problem-solving frameworks, and quantitative deliverables behind key civil, bridge, geotechnical, and BIM projects completed by Zaidan Dzaki Prasetya Adi. Structured in high-level consulting case study formats (Project Brief, Role & Technology Stack, Technical Challenges, and Engineering Solutions & Quantitative Impacts), it highlights technical competence across Indonesian National Standards (SNI), advanced finite element modeling, and field constructability.

Case Study 01: Aik Bari IV Cable-Stayed Bridge Redesign (240 m)

Category: Long-Span Bridges / Bachelor Degree Capstone Project **Period:** Jun 2025 – Jan 2026

Client / Academic Scope: Department of Civil Engineering, Institut Teknologi Sepuluh Nopember (ITS)

1. Project Brief:

Comprehensive structural redesign and geometric modification of the 240-meter span Aik Bari IV Bridge, transforming an existing multi-span steel truss into a signature, high-performance semi-harp cable-stayed bridge. The objective was to increase navigation clearance, optimize life-cycle durability, and improve structural aerodynamic behavior under severe wind and seismic excitations.

2. My Role & Technology Stack:

Role: Lead Structural Analyst & BIM Modeler

Tools: MIDAS Civil (Construction Stage & Cable Tuning), SAP2000 (Global Dynamic Analysis), Autodesk Revit (LOD 350 BIM Model), AutoCAD, SMath Studio (Analytical Verification Worksheets).

3. Technical Challenges:

- **Non-linear Cable-Structure Interaction & Deck Deflection:** Managing geometric non-linearities ($P-\Delta$, large displacement, cable sag effect) and coupling forces between high-tensile stay cables and an orthotropic steel box girder during progressive balanced cantilever erection.
- **Seismic & Aerodynamic Flutter Risk:** Mitigating high dynamic excitation and flutter vulnerability across the 240 m flexible navigation span under high seismic PGA zone conditions in Indonesia.

4. Engineering Solutions & Quantitative Deliverables:

- **Initial Tension Optimization & Construction Staging:** Performed backward iteration and Unknown Load Factor (ULF) optimization in MIDAS Civil to determine precise initial cable prestressing forces, guaranteeing near-zero bending moment in the concrete pylon and zero-tension stress states in the deck upon bridge closure.
- **High-Efficiency Aerodynamic Section:** Selected and engineered an aerodynamic orthotropic steel box girder profile with semi-harp stay geometry, increasing torsional stiffness and satisfying aeroelastic flutter velocity criteria per SNI 2833:2016.
- **Full BIM Integration (LOD 350):** Generated parametric 3D structural BIM models in Autodesk Revit encompassing anchor heads, shear connectors, stay anchorages, sub-structures, and foundation bored piles, delivering clash-free shop-level documentation.

Case Study 02: Integrated BIM Modeling for 10-Storey Building & Civil Infrastructure

Category: Building Information Modeling (BIM) & Clash Resolution **Period:** May 2024 – Feb 2026

Scope: Multi-Disciplinary Digital Engineering & 4D Coordination

1. Project Brief:

End-to-end digital engineering and Level of Development (LOD) 300–350 BIM implementation for a complex 10-storey reinforced concrete mixed-use facility as well as specialized civil bridge infrastructure (tied-arch and truss networks). The engagement focused on transitioning 2D conceptual designs into coordinated, constructible digital twin assets.

2. My Role & Technology Stack:

Role: Structural BIM Modeler & Interdisciplinary Coordinator

Tools: Autodesk Revit (Structural, Architectural & Rebar), Autodesk Navisworks Manage, AutoCAD, Lumion 3D.

3. Technical Challenges:

- **Spatial Bottlenecks & Severe Clashes:** Resolving multi-trade spatial conflicts between heavily reinforced moment-resisting transfer beams, shear walls, vertical MEP risers, and drainage lines within confined ceiling plenums.
- **Model Fidelity vs. Performance:** Maintaining smooth model performance while generating fully detailed parametric reinforcement, complex connection plates, and high-density geometric families.

4. Engineering Solutions & Quantitative Deliverables:

- **Automated Clash Detection Matrix:** Executed comprehensive hard/soft clash audits in Navisworks Manage, detecting and resolving **50+ interdisciplinary design collisions** prior to tender release, preventing costly on-site change orders.
- **High-Precision Quantity Take-Off (QTO):** Leveraged Revit's scheduling engine to extract dynamic material schedules for concrete pours, structural steel tonnage, and rebar weights, attaining **98% volumetric alignment** with final Bills of Quantities (BOQ).
- **Parametric Family Formulation:** Engineered reusable parametric families for structural columns, haunched beams, elastomeric bearings, and bridge abutments, accelerating cross-project drafting productivity by over 30%.

Case Study 03: Muhammadiyah Babat Hospital (8-Storey RC Construction)

Category: Site Supervision, QA/QC & Constructability Engineering **Period:** Dec 2024 – Feb 2025

Project Location: Babat, Lamongan, East Java

1. Project Brief:

On-site technical supervision, quality assurance/quality control (QA/QC), and constructability management during the structural erection of an 8-storey reinforced concrete hospital facility designed with Special Moment Resisting Frames (SRPMK).

2. My Role & Technology Stack:

Role: Site Engineering Intern & Assistant QA/QC Supervisor

Tools: Autodesk Revit (Bar Bending Schedules & QTO), MS Excel, Slump Cone, Compression Testing Apparatus.

3. Technical Challenges:

- **Congested Seismic Beam-Column Joint Detailing:** Managing high steel congestion at beam-column joint cores mandated by SNI 2847:2019 SRPMK provisions without triggering honeycomb defects during concrete placement.
- **Material Waste & Tight Staging Yards:** Rebar fabrication cutting waste and logistics bottlenecks within an active hospital expansion zone with limited on-site storage.

4. Engineering Solutions & Quantitative Deliverables:

- **BIM-Driven Rebar Optimization (BBS):** Developed precise 3D Bar Bending Schedules (BBS) and cutting lists in Autodesk Revit, slashing rebar off-cut scrap rates by approximately **7%** and preventing bar placement clashes.
- **Rigorous Concrete QA/QC Protocol:** Supervised sampling, slump testing, and standard compressive cylinder crush tests (7, 14, and 28 days) per ASTM C39 / SNI 2847 standards, verifying 100% design strength compliance prior to formwork stripping.
- **Productivity Cycle Time Monitoring:** Evaluated tower crane usage cycles and concrete pouring productivity rates, formulating critical-path progress reports that safeguarded schedule milestones.

Case Study 04: Earth Retaining Structures, Slope Stability & Deep Foundations

Category: Geotechnical Design, Forensic Soil Mechanics & Modeling **Period:** Aug 2024 – Jun 2025

Scope: Major Geotechnical Capstone Studies & Site Investigation Projects

1. Project Brief:

Geotechnical engineering design, global slope stability analysis, and foundation design for critical civil infrastructure subjected to lateral earth pressures, high groundwater tables, and heavy live surcharges. Included comparative evaluations of retaining walls, sheet piles, and deep bored pile groups.

2. My Role & Technology Stack:

Role: Geotechnical Analysis & Design Modeler

Tools: GEO5 (Sheeting Design, Cantilever Wall, Slope Stability, Pile Group), SMath Studio, AutoCAD, Soil Lab Testing Devices.

3. Technical Challenges:

- **Soft Alluvial Soil Layers & Basal Heave:** Low undrained shear strength and high compressibility in alluvial soils posing imminent risks of deep rotational slope failure and excessive lateral wall deflection.
- **Seismic Soil-Structure Surcharge:** Incorporating pseudo-static seismic coefficients into earth pressure envelopes without oversizing retaining wall cross-sections.

4. Engineering Solutions & Quantitative Deliverables:

- **Comprehensive Stability Verification in GEO5:** Evaluated multiple failure mechanisms (sliding, overturning, bearing capacity, and overall circular slip using Bishop & Fellenius methods), confirming Safety Factors exceeded code requirements ($SF_{static} \geq 1.50$, $SF_{seismic} \geq 1.10$).
- **Comparative Structural Assessment:** Formulated an engineering trade-off matrix comparing gravity RC walls, anchored cantilever sheet piles, and geotextile-reinforced slopes, identifying the sheet pile system with prestressed ground anchors as the optimal solution that kept crest deflection below ≤ 25 mm.
- **Deep Foundation Capacity Analysis:** Calculated individual pile and pile group ultimate axial bearing capacities (using Meyerhof, Vesic, and alpha-beta methods) and lateral resistance curves based on in-situ CPT and SPT borehole profiles.

Case Study 05: Earthquake Engineering & Structural Bridge Competitions

Category: High-Performance Seismic Design & Bridge Innovation Awards **Period:** 2023 – 2025

Venues: IDEERS Taiwan, CIVEX, CIVEST, CIA, CREATION UGM

1. Project Brief:

Design, dynamic structural modeling, and physical prototyping of innovative structural configurations under severe dynamic seismic loads (up to 800 Gal) and long-span bridge loading conditions.

2. My Role & Technology Stack:

Role: Lead Structural Modeler & Team Analyst

Tools: SAP2000, MIDAS Civil, SMath Studio, Physical Shake Table & Dial Gauge Calibration Jigs.

3. Engineering Highlights & Achievements:

- **IDEERS 2025 (Taiwan) — 8th Place & Innovation Award:** Engineered a 60 cm seismic model fitted with custom rubber base-isolation dampers. Sustained shake-table testing up to **800 Gal acceleration** out of 49 international university teams through controlled plastic hinging and energy dissipation.
- **CIVEX 2024 — 1st Place Winner & Favorite Award:** Modeled and presented a 200 m pedestrian tied-arch bridge optimizing cable hanger tensions to minimize localized bending moments in the arch ribs.
- **CIVEST 2025 — 2nd Runner-Up:** Engineered “Rajansa Bridge,” a double-span tied-arch steel bridge (2×80 m), optimizing steel weight-to-load capacity efficiency by 15%.

Document Note: This engineering dossier serves as the technical companion to the visual portfolio presentation. Full calculation spreadsheets, finite element input files, and high-resolution BIM models are available upon request.